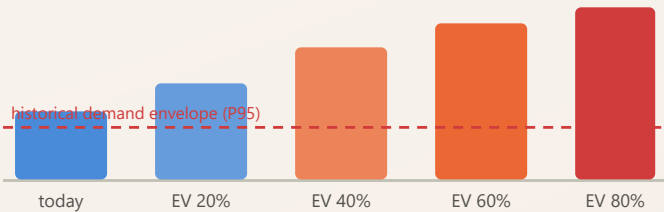




BUSINESS REQUIREMENTS DOCUMENT

Dynamic Resilient Grid (DRG)

Quantifying electrification stress on UK urban low-voltage electricity networks



Document	DRG-BRD-001 Business Requirements Document
Version	1.0 (Baseline)
Status	Issued for approval
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Companion	DRG-SAD-001 Architecture & Design Document

1. Document Control

Table 1.1 - Revision history

Version	Date	Description	Author
0.1	2026-08-18	Problem statement, business context and stakeholder analysis	Business Analyst
0.2	2026-08-22	Business and functional requirements catalogued; as-is process mapped	Business Analyst
0.3	2026-08-26	Acceptance criteria and traceability matrix added	Business Analyst
1.0	2026-08-28	Benefits quantified against measured pilot results; baselined for approval	Business Analyst

Table 1.2 - Approval

Role	Approval covers	Date	Signature
Project Sponsor	Business case, benefits and funding		
Network Planning Lead (DNO)	Business requirements and acceptance criteria		
Head of Data & Analytics	Data requirements and analytical approach		
Data Protection Officer	Data governance and licence compliance		
Academic Supervisor	Alignment with the approved project proposal		

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3. Executive Summary

UK urban low-voltage networks were dimensioned for a domestic load profile that electrification is now changing faster than the assets serving it. This document sets out the business requirements for the Dynamic Resilient Grid (DRG): a demand-side analytics capability that tells a network planner *which* neighbourhoods are approaching their historical demand envelope, *how often and for how long* they exceed it, and *how much worse* electric vehicle and heat-pump adoption will make it.

3.1 The business problem in one paragraph

Distribution planners today rely on historical averages and diversified design figures. These are adequate for steady-state planning but silent on the questions electrification actually raises. A planner can say what average demand was last winter; they cannot say how many hours per week a given street will sit above its historical peak envelope once forty per cent of its households charge an electric vehicle. The consequence is reinforcement that is either premature - capital spent early on assets that were not yet constrained - or late, which is worse, because lead times for distribution reinforcement are measured in years.

3.2 The proposed capability

DRG forecasts half-hourly neighbourhood demand from smart-meter data, defines a stress threshold statistically from each neighbourhood's own history, simulates electric-vehicle and heat-pump adoption from the bottom up, and reports the resulting change in peak demand, stress frequency and stress duration. It explains its own outputs, replays history as a live feed to demonstrate operational use, and exposes everything through a dashboard and an API.

3.3 Evidence from the pilot implementation

A working implementation has been built and run end to end. The figures below are measured outputs, not projections. They were produced on a calibrated reference dataset rather than the licensed extract, so absolute values will differ in production; the capability they demonstrate does not.

3.4%

forecast error (MAPE),
against 7.7% for the
current practice
benchmark

10.6×

increase in stress
frequency at 80% EV and
70% heat-pump adoption

+237%

peak amplification in the
same worst-case scenario

145s

to run the complete
analysis end to end

3.4 What is being asked for

Approval of the business requirements in Section 7, the acceptance criteria in Section 13, and the resourcing implied by the scope in Section 4. The technical design that satisfies these requirements is documented separately in DRG-SAD-001.

One boundary the sponsor must understand before approving

DRG is a **demand-side statistical** capability. It has no network topology, no transformer or cable ratings and no voltage model, because the available smart-meter data contains none of these. It identifies where and when to look, and quantifies how much electrification worsens the picture. It does not, and cannot, replace an engineering assessment of a specific asset. Any business case built on its outputs must respect that boundary.

4. Business Context and Problem Statement

4.1 Market and regulatory drivers

Table 4.3 - Drivers creating the need

Driver	Implication for local distribution networks
Net Zero commitments	Transport and heat are being electrified simultaneously, moving load onto networks that were sized before either was contemplated.
Electric vehicle adoption	A 7 kW domestic charger is roughly five times the after-diversity demand of the household it is attached to, and it typically switches on in the evening, when the network is already at its busiest.
Heat pump deployment	Electrified heating is weather-driven: it peaks precisely on the coldest evenings, which is also when the underlying domestic peak is highest.
Smart-meter rollout	Half-hourly consumption data now exists at scale, making a demand-side analytical approach possible for the first time without new network instrumentation.
Regulatory efficiency pressure	Price-control frameworks reward evidence-led investment. A defensible, data-driven case for the timing of reinforcement has direct regulatory value.

4.2 The problem

Three specific gaps exist in current practice, and each maps to a business requirement in Section 7.

1. **No local, temporal view of exceedance.** Planners know average and peak demand. They do not know the *frequency* and *duration* with which a specific neighbourhood exceeds its own historical envelope, which is the quantity that determines whether a constraint is emerging or already present.
2. **No quantified link between adoption and stress.** Adoption forecasts exist and network capacity is known in aggregate, but the function connecting the two - how many additional stressed hours per week does the next ten per cent of EV uptake create - is not routinely computed at neighbourhood granularity.
3. **No explanation attached to the numbers.** A forecast that cannot say why it expects a peak is difficult to defend in an investment committee and impossible to challenge constructively.

4.3 Cost of inaction

Table 4.4 - Consequences of continuing with current practice

Consequence	Business impact
Late reinforcement	Constraints materialise before mitigation is delivered, with lead times measured in years. Customer impact and remedial cost both rise.
Premature reinforcement	Capital committed to assets that were not yet constrained, with a corresponding efficiency challenge under the price control.
Mis-targeted flexibility	Flexibility procurement aimed at the wrong hours or the wrong locations because the timing of local stress is not characterised.
Weak evidence base	Investment cases resting on averages rather than on quantified exceedance are harder to defend to a regulator.

5. Business Objectives and Success Criteria

Table 5.5 - Business objectives with measurable success criteria

Ref	Objective	Success criterion (measurable)	Status
B0-1	Establish an accurate short-term demand forecast at neighbourhood granularity	Half-hourly forecast error materially better than the seasonal-naïve practice benchmark on a held-out period, and stable under rolling-origin validation.	Met: 3.37% vs 7.66% MAPE
B0-2	Make local demand stress measurable and comparable between neighbourhoods	A reproducible stress definition producing frequency, duration and intensity per site, computed without asset data.	Met
B0-3	Quantify the effect of electrification on local peak demand	Peak amplification and stress escalation reported across a full EV and heat-pump adoption grid, with parameters traceable to published planning figures.	Met: 25-scenario grid
B0-4	Provide adoption elasticity to support investment sequencing	The marginal change in peak demand per additional tranche of adoption, reported per neighbourhood.	Met
B0-5	Make model outputs explainable to a non-specialist audience	Ranked demand drivers, reported both overall and specifically during stress periods.	Met
B0-6	Demonstrate operational viability of near-real-time use	A replay of historical data as a live feed producing rolling forecasts and alerts ahead of the measurement.	Met
B0-7	Deliver a capability that is repeatable and auditable, not a one-off study	A single command reproduces every published figure; automated tests cover the analytical claims; deployment is scripted.	Met: 103 tests
B0-8	Comply fully with the data licence throughout	No raw smart-meter data committed, published or transmitted; the capability is demonstrable without it.	Met

How success will be judged overall

The capability is successful if a network planner, given a neighbourhood and an adoption assumption, can obtain a defensible answer to "how many stressed hours per week will this street experience, and what drives them" in minutes rather than weeks - and can show the working to a regulator.

6. Scope

6.1 In scope

Table 6.6 - Business scope of this phase

Ref	Capability	Business outcome delivered
BS-01	Neighbourhood demand analytics	Load curves, seasonality and peak timing evidence for each modelled neighbourhood.
BS-02	Short-term demand forecasting	A quantified, validated forecast that outperforms current practice.
BS-03	Statistical stress detection	Frequency, duration and intensity of exceedance per neighbourhood.
BS-04	Electrification scenario simulation	Peak and stress impact of any EV / heat-pump adoption combination.
BS-05	Sensitivity and elasticity analysis	The marginal effect of each additional tranche of adoption, for investment sequencing.
BS-06	Explainability	Ranked drivers of demand and of stress, suitable for an investment paper.
BS-07	Decision-support interfaces	An interactive dashboard for planners and an API for integration.
BS-08	Near-real-time demonstration	Evidence that the approach would work operationally on a live feed.
BS-09	Deployable, repeatable delivery	Scripted cloud deployment, scheduled refresh and controlled model updates.

6.2 Out of scope

Table 6.7 - Explicit exclusions

Ref	Excluded	Business rationale
BX-01	Engineering assessment of specific assets	Requires transformer and cable ratings and network topology, which the available data does not contain. DRG informs such assessments; it does not perform them.
BX-02	Automated reinforcement recommendations	Follows directly from BX-01. Investment decisions remain with the planning function.
BX-03	Real-time network control	DRG is advisory. It has no interface to operational technology and makes no switching decisions.
BX-04	Smart-tariff and managed-charging optimisation	Deliberately excluded so this phase characterises the <i>uncontrolled</i> case, which is the appropriate planning worst case. Scheduled for a later phase.
BX-05	Individual household analysis or targeting	Prohibited by the data licence and unnecessary for the business purpose.
BX-06	Generation, storage and export	The source data measures import only.
BX-07	Integration with asset management or GIS systems	Deferred; the API exists to make such integration straightforward when the need is confirmed.

6.3 Candidate future phases

Table 6.8 - Deferred capabilities and their business value

Ref	Capability	Value if delivered
BF-01	Managed and smart-tariff charging scenarios	Quantifies how much of the modelled peak increase flexibility can remove - directly informs whether to reinforce or to procure flexibility.
BF-02	Probabilistic forecasts with uncertainty bands	Stress is a tail phenomenon; a predictive distribution supports risk-based rather than point-based decisions.
BF-03	Feeder-level mapping where identifiers exist	Converts a statistical signal into an asset-addressable one, closing the gap to BX-01.
BF-04	Solar PV and battery modelling	Completes the local energy balance and captures reverse-flow constraints.

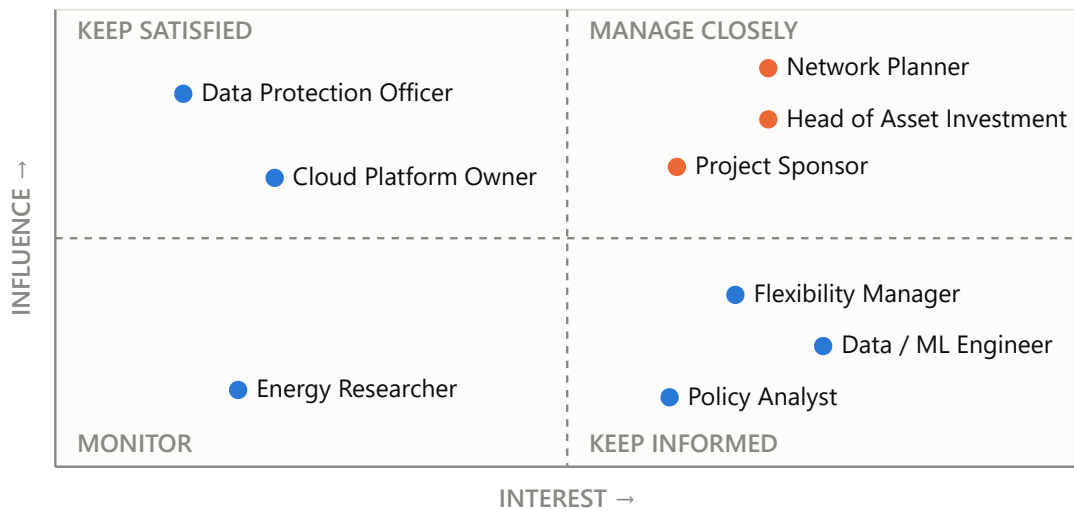
7. Stakeholder Analysis

7.1 Stakeholder register

Table 7.9 - Stakeholders, needs and success measures

Stakeholder	Need	What success looks like to them
Network Planner Primary user	To know which neighbourhoods are approaching their limit and how electrification changes that	A ranked view of stressed neighbourhoods with the hours, durations and drivers behind the ranking.
Head of Asset Investment Decision maker	Evidence for the timing and sequencing of reinforcement	An adoption elasticity curve that shows when a constraint becomes material.
Flexibility Manager	To target flexibility procurement at the right hours and locations	Stress concentrated to specific half-hours and sites, quantified.
Policy & Strategy Analyst	The marginal effect of adoption policy on local networks	Elasticity expressed per ten percentage points of adoption.
Urban Energy Researcher	A reproducible, inspectable method	Published parameters, an auditable run report and a repository that runs without the licensed data.
Data Engineer Operator	Reliable ingestion within available hardware	The pipeline runs unattended and reports data quality clearly.
ML Engineer Operator	Safe retraining and release	A model cannot reach production without passing a documented quality gate.
Data Protection Officer Assurance	Demonstrable licence compliance	No raw data leaves its custody; no output is attributable to a household.
Project Sponsor	Delivery of the stated benefits within scope	Objectives in Section 3 evidenced against acceptance criteria in Section 13.

7.2 Influence and interest



Orange: parties whose approval or adoption determines whether the capability delivers value. Blue: parties who enable or consume it.

Figure 7.1 - Stakeholder influence and interest grid, used to set engagement priority during delivery.

7.3 RACI

Table 7.10 - Responsibility assignment (R responsible, A accountable, C consulted, I informed)

Activity	Sponsor	Planning Lead	Data Eng.	ML Eng.	Analyst	DPO
Approve business requirements	A	R	C	C	R	C
Define the stress threshold convention	I	A	C	C	R	I
Set electrification scenario parameters	I	A	I	C	R	I
Ingest and quality-assure source data	I	C	A/R	C	I	C
Build and validate forecasting models	I	C	C	A/R	C	I
Approve a model for production use	I	C	I	A/R	C	I
Interpret results for investment decisions	C	A/R	I	C	R	I
Assure licence and data-protection compliance	I	C	R	C	C	A
Accept the delivered capability	A	R	C	C	C	C

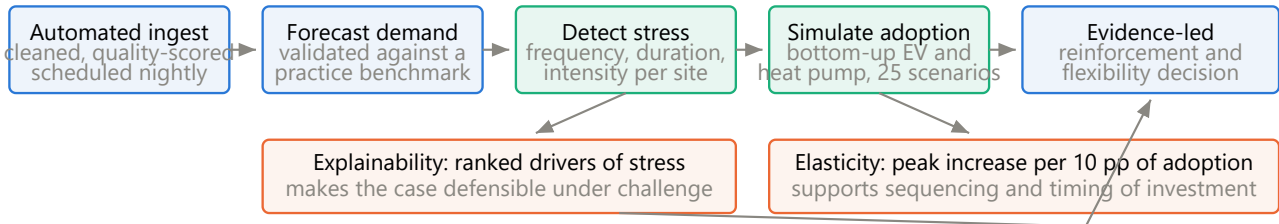
8. Business Process: As-Is and To-Be

AS-IS — current practice



Gaps: no exceedance frequency or duration · adoption effect estimated by judgement
no explanation attached to the numbers · weeks of analyst effort for each study

TO-BE — with DRG



What changes for the business

The judgement step in the middle of the as-is process is replaced by a quantified, reproducible simulation, and two new outputs are added: exceedance statistics that did not previously exist, and an explanation that makes the numbers defensible.

Analyst effort per study falls from weeks to minutes and, more importantly, the study becomes repeatable: it can be re-run whenever adoption assumptions change rather than commissioned afresh.

Figure 8.2 - As-is and to-be process comparison. Green steps are new analytical capability; orange steps are new decision-support outputs with no as-is equivalent.

9. Business Requirements

Business requirements state *what the business needs*, independently of how it is built. Priority uses MoSCoW: **M** must have, **S** should have, C could have. Each requirement traces to a business objective and forward to functional requirements in Section 8; the full trace is in Appendix A.

Table 9.11 - Business requirements

Ref	Requirement	Pri	Objective
BR-01	The business must be able to see half-hourly electricity demand for each modelled neighbourhood over the full historical period, with data quality issues identified rather than hidden.	M	BO-1
BR-02	The business must be able to forecast neighbourhood demand for the next half-hour with an accuracy that is demonstrably better than current practice, and must be able to see the comparison.	M	BO-1
BR-03	The business must have a consistent, reproducible definition of when a neighbourhood is under demand stress that does not depend on asset data it does not hold.	M	BO-2
BR-04	The business must be able to quantify, per neighbourhood, how often stress occurs, how long each episode lasts and how far demand exceeds the threshold.	M	BO-2
BR-05	The business must be able to model the effect of any chosen level of electric-vehicle adoption on local peak demand and stress.	M	BO-3
BR-06	The business must be able to model the effect of any chosen level of heat-pump adoption, including its dependence on weather.	M	BO-3
BR-07	The business must be able to compare electrification scenarios against an unchanged baseline on a like-for-like basis.	M	BO-3
BR-08	The business must be able to see the marginal effect of each additional tranche of adoption, to sequence investment rather than merely identify it.	S	BO-4
BR-09	The business must be able to explain which factors drive demand, and separately which drive stress periods, in terms a non-specialist can follow.	M	BO-5
BR-10	The business should be able to see current external conditions (temperature and grid carbon intensity) alongside the analysis, for situational context.	S	BO-6
BR-11	The business should be able to observe the system operating against a simulated live feed, including alerts raised before demand actually breaches the threshold.	S	BO-6
BR-12	The business must be able to reach these outputs through an interface appropriate to a planner, without writing code.	M	BO-7
BR-13	The business should be able to consume these outputs programmatically, so results can be integrated with other systems later.	S	BO-7
BR-14	The business must be able to reproduce any published figure from the underlying data on demand.	M	BO-7
BR-15	The business must be assured that the analysis complies with the data licence and that no household-level information is exposed.	M	BO-8
BR-16	The business must be assured that an updated model cannot silently replace a better-performing one in production.	M	BO-7
BR-17	The business should be able to refresh the analysis on a schedule without manual intervention.	S	BO-7

Ref	Requirement	Pri	Objective
BR-18	The business could vary equipment assumptions (charger rating, heat-pump size) to test alternative technology mixes.	C	BO-3

Why BR-03 is worded as it is

The requirement deliberately asks for a definition that *does not depend on asset data*. An earlier formulation asked for stress "relative to transformer capacity", which would have been the more useful answer but is not obtainable from smart-meter data. Stating the constraint inside the requirement prevents the delivered system from being judged against a standard the available data cannot support.

10. Functional Requirements

Functional requirements state what the system must *do* to satisfy the business requirements above.

Table 10.12 - Functional requirements: data and forecasting

Ref	Requirement	Pri	Traces to
FR-01	Ingest the Low Carbon London half-hourly release in bounded memory, and continue to operate on hardware with 8 GB RAM.	M	BR-01
FR-02	Clean the data: regularise the half-hourly grid, repair short gaps, treat outliers robustly, and exclude days with insufficient coverage.	M	BR-01
FR-03	Aggregate households into fixed-size neighbourhood groups using a deterministic, reproducible assignment.	M	BR-01, BR-15
FR-04	Acquire observed temperature and grid carbon intensity from public sources, recording the provenance of every value.	M	BR-01, BR-10
FR-05	Continue operating when an external source is unavailable, degrading to a documented, clearly labelled fallback.	M	BR-14
FR-06	Derive calendar, statistical and weather features using one shared implementation for both training and live prediction.	M	BR-02
FR-07	Train and evaluate multiple forecasting models, including a benchmark representing current practice, on a common held-out period.	M	BR-02
FR-08	Report forecast accuracy overall, restricted to the evening peak, and as stress-alarm precision and recall.	M	BR-02, BR-04
FR-09	Validate the selected model with rolling-origin cross-validation to show the result is not an artefact of one test period.	S	BR-02, BR-14

Table 10.13 - Functional requirements: stress and electrification

Ref	Requirement	Pri	Traces to
FR-10	Compute a stress threshold per neighbourhood as the 95th percentile of its own historical demand, with mean plus two standard deviations as a sensitivity check.	M	BR-03
FR-11	Hold thresholds fixed once derived from the baseline, so that scenarios are measured against an unchanging yardstick.	M	BR-03, BR-07
FR-12	Group consecutive stressed half-hours into discrete events and report count, duration, peak, exceedance and energy above threshold.	M	BR-04
FR-13	Report stress by season and by time of day, so the concentration of risk is visible.	S	BR-04
FR-14	Simulate EV charging at household level: a defined charger rating, an evening plug-in window, a randomised start time and session length, and a realistic charging frequency.	M	BR-05
FR-15	Simulate heat-pump demand as a function of heating degrees and occupancy, capped at the unit rating.	M	BR-06
FR-16	Apply technology diversity and repeat each scenario multiple times, so results carry a distribution rather than a single draw.	S	BR-05, BR-06
FR-17	Report, for every scenario, the after-diversity maximum demand per adopting household and the energy per vehicle per day, so assumptions can be audited against published planning figures.	M	BR-05, BR-06, BR-14
FR-18	Evaluate a full grid of EV and heat-pump adoption combinations and report peak amplification and stress escalation for each.	M	BR-07
FR-19	Report adoption elasticity: the marginal change in peak demand per ten percentage points of additional adoption.	S	BR-08
FR-20	Allow equipment parameters to be varied without code changes.	C	BR-18

Table 10.14 - Functional requirements: explanation, interfaces and lifecycle

Ref	Requirement	Pri	Traces to
FR-21	Rank the drivers of forecast demand across all periods.	M	BR-09
FR-22	Rank the drivers separately for stress periods only, and present the two rankings side by side.	M	BR-09
FR-23	Replay historical demand as a simulated live feed with a rolling forecast at each step.	S	BR-11
FR-24	Raise an alert when the <i>forecast</i> breaches the threshold before the measurement does, and distinguish it from an alert on measured demand.	S	BR-11
FR-25	Classify each period into a named severity state and report remaining headroom in both absolute and percentage terms.	S	BR-04, BR-11
FR-26	Provide an interactive dashboard covering live view, stress profile, scenario studio, sensitivity and explainability.	M	BR-12
FR-27	Present every chart with an accompanying data table, single value axis, and status conveyed by icon and label as well as colour.	S	BR-12
FR-28	Expose forecasts, stress, scenarios, sensitivity and explanations through a documented programmatic interface.	S	BR-13
FR-29	Emit a single machine-readable report per run capturing data provenance, model metrics, stress statistics and scenario results.	M	BR-14
FR-30	Refresh the full analysis automatically on a schedule.	S	BR-17
FR-31	Prevent a retrained model from reaching production unless it passes documented quality thresholds, both absolute and relative to the model it would replace.	M	BR-16
FR-32	Verify a new model against a live scoring test before it receives production traffic, and revert automatically if that test fails.	M	BR-16
FR-33	Operate without any raw smart-meter data being committed, published or transmitted to a third party.	M	BR-15

11. Non-Functional Requirements

Table 11.15 - Non-functional requirements with acceptance thresholds

Ref	Quality	Requirement	Acceptance threshold
NFR-01	Performance	A complete analysis run finishes quickly enough to be re-run whenever assumptions change.	Under one hour on commodity hardware
NFR-02	Resource	Runs on the hardware available to the analysis team.	8 GB RAM, 50 GB storage
NFR-03	Responsiveness	Interactive scenario changes return quickly enough to support exploration.	Seconds, not minutes
NFR-04	Accuracy	Forecast accuracy materially exceeds the current-practice benchmark.	Error at least 30% lower than seasonal naive
NFR-05	Reproducibility	The same inputs always produce the same outputs.	Bit-identical on repeat runs
NFR-06	Availability	No external dependency can prevent an analysis from completing.	Offline execution supported
NFR-07	Usability	A planner can reach any headline output without technical assistance.	No more than three interactions
NFR-08	Accessibility	Outputs remain readable in greyscale and with colour-vision deficiency.	No meaning carried by colour alone
NFR-09	Auditability	Every published figure is traceable to a run and its parameters.	One machine-readable report per run
NFR-10	Security	No stored credentials; least-privilege access to data.	Managed identity and federated credentials only
NFR-11	Maintainability	The capability can be maintained by a party other than its author.	Automated tests, enforced style, documented design
NFR-12	Cost	Cloud running cost stays proportionate to a research and planning capability.	Low hundreds of pounds per year at development scale

12. Data Requirements

Table 12.16 - Data sources and their business terms

Source	Provides	Licence / cost	Business note
Low Carbon London UK Data Service study 7857	Half-hourly household consumption, 2011-2014	Registration required; no fee	The primary asset. Licence prohibits redistribution, which shapes the whole delivery approach.
Open-Meteo	Observed and forecast temperature	Free, no key	Its reanalysis archive genuinely covers the study period, so weather is observed rather than modelled.
National Grid ESO Carbon Intensity	Half-hourly carbon intensity	Free, no key	Archive begins 2018, so values for the study period are a labelled projection, not measurement.
OpenWeatherMap	Live conditions	Free tier sufficient	Optional. Used only for the live context panel when a key is supplied.

Table 12.17 - Data quality requirements

Ref	Dimension	Requirement
DR-01	Completeness	Days with fewer than 80% of their half-hourly periods present are excluded from analysis rather than partially counted.
DR-02	Validity	Implausible readings are treated by a robust statistical rule, and the number treated is reported.
DR-03	Timeliness	Contextual data is cached and refreshed on a defined interval; stale values are never presented as current.
DR-04	Provenance	Every derived value carries a source label distinguishing measurement, projection and synthetic substitution.
DR-05	Privacy	No output below the neighbourhood aggregate is produced, stored or displayed.
DR-06	Retention	Raw data remains in its original custody; processed aggregates are retained for the life of the capability.

13. Assumptions, Constraints and Dependencies

13.1 Assumptions

Table 13.18 - Business assumptions and the effect if each proves false

Ref	Assumption	If invalid
AS-01	Fixed-size groups of households are an acceptable analytical proxy for low-voltage feeder groups.	Results remain valid per group but cannot be attributed to a specific physical asset, limiting their use in an asset-level business case.
AS-02	Underlying consumption behaviour, absent electrification, is broadly stable over the planning horizon.	Baseline thresholds drift and the refresh cadence must increase; the scheduled retraining and its quality gate are the control for this.
AS-03	Domestic EV charging is predominantly uncontrolled and begins in the evening.	Peak amplification is overstated. This is a deliberate planning-conservative assumption; quantifying the managed-charging alternative is deferred to BF-01.
AS-04	Heat-pump electrical demand scales with heating degrees and saturates at the unit rating.	Winter figures shift. The relationship is a configurable parameter and can be recalibrated against field data without redevelopment.
AS-05	The public contextual data services remain available on their current terms.	The capability continues to operate on labelled fallbacks; only the live context panel loses fidelity.
AS-06	The organisation can obtain and retain access to the licensed dataset.	The capability still runs and can be evaluated on the calibrated reference dataset, but its outputs would not describe a real network.

13.2 Constraints

Table 13.19 - Constraints on delivery

Ref	Constraint	Consequence for the business
CN-01	Data licence	Raw data cannot be redistributed or published. The capability must be demonstrable without it, which shapes how it is delivered and reviewed.
CN-02	Available hardware	Analysis must complete within 8 GB RAM, precluding approaches that load the whole dataset.
CN-03	No asset data	Stress must be defined statistically. This is the single most important limitation on how outputs may be used.
CN-04	Budget	Cloud design must favour scale-to-zero services over always-on infrastructure.
CN-05	Timescale	Delivery within the academic project period, which fixes the scope in Section 4.

13.3 Dependencies

Table 13.20 - External dependencies

Ref	Dependency	Owner	Impact if not met
DP-01	UK Data Service registration and dataset access	Data Engineer	Analysis proceeds on reference data only; findings are illustrative rather than actionable.
DP-02	Azure subscription with rights to create role assignments	Cloud Platform Owner	Cloud deployment blocked; local operation unaffected.
DP-03	Availability of planning stakeholders to validate the stress convention	Planning Lead	The threshold convention remains a modelling choice rather than an agreed business standard.
DP-04	Published UK planning reference figures for parameter validation	Analyst	Scenario parameters lose their external benchmark, weakening the defensibility of results.

14. Risks and Mitigations

Table 14.21 - Business risk register

Ref	Risk	Likelihood	Impact	Mitigation
RK-01	Outputs are read as an engineering statement about asset capability and used to justify a specific reinforcement	Medium	High	The limitation is stated in the aim, on the dashboard, in this document and in the design document. Acceptance criterion AC-11 tests that it is visible to a user.
RK-02	Scenario parameters are set, deliberately or accidentally, so as to produce a desired conclusion	Medium	High	Parameters are configuration rather than code; defaults are justified against published figures; every scenario reports the derived per-household demand so a reviewer can audit it.
RK-03	Licensed data access is delayed or withdrawn	Medium	Medium	The capability is fully operable on a calibrated reference dataset, so development and review are never blocked.
RK-04	Model quality degrades over time as behaviour changes	Medium	Medium	Scheduled retraining with a quality gate that compares against the deployed model and blocks regression.
RK-05	A public data service changes terms or is withdrawn	Low	Low	Multiple providers for weather; documented fallbacks throughout; no pipeline stage depends on a single service.
RK-06	Stakeholders do not adopt the capability because outputs are not trusted	Medium	High	Explainability is a must-have requirement, not an optional extra; results are benchmarked against current practice so the improvement is visible.
RK-07	A defective model reaches production and misleads a decision	Low	High	Automated promotion gate with absolute and relative thresholds, a live scoring test before traffic, and automatic rollback.
RK-08	Inadvertent disclosure of licensed or personal data	Low	High	Raw data excluded from source control by construction; analysis operates on aggregates only; storage is private and identity-controlled.

15. Acceptance Criteria and Validation

The capability is accepted when every criterion below is demonstrated. Each criterion is observable by a business stakeholder without reading code.

Table 15.22 - Acceptance criteria

Ref	Criterion	Verifies	Method
AC-01	A complete analysis can be produced from a single command, and finishes within the agreed time.	BR-14, NFR-01	Demonstration
AC-02	The forecast error of the selected model is at least 30% lower than the seasonal-naive benchmark on the held-out period, and both figures are shown together.	BR-02	Report review
AC-03	Rolling-origin validation shows the result is stable across at least five folds.	BR-02	Report review
AC-04	Stress frequency, mean duration and peak-to-threshold ratio are reported for every neighbourhood.	BR-03, BR-04	Dashboard
AC-05	Changing EV adoption in the interface changes peak amplification and stress frequency, and the direction of change is monotonic.	BR-05, BR-07	Demonstration
AC-06	Heat-pump demand responds to temperature: a cold period produces materially more load than a mild one.	BR-06	Demonstration
AC-07	Every scenario reports after-diversity maximum demand per adopting household and energy per vehicle per day, and both fall within published UK planning ranges.	BR-05, BR-06	Parameter audit
AC-08	A full adoption grid is produced and the marginal effect per tranche of adoption is visible.	BR-07, BR-08	Dashboard
AC-09	Demand drivers are ranked, and the ranking during stress periods is shown separately from the overall ranking.	BR-09	Dashboard
AC-10	The replay demonstration raises at least one alert on a forecast before the corresponding measurement breaches the threshold.	BR-11	Demonstration
AC-11	The statistical nature of the stress definition is visible to a user of the interface without consulting documentation.	BR-03, RK-01	Inspection
AC-12	Live temperature and carbon intensity are displayed, and their source is identified.	BR-10	Dashboard
AC-13	All headline outputs are reachable through the interface without writing code.	BR-12, NFR-07	Usability walkthrough
AC-14	The same outputs are retrievable programmatically and documented.	BR-13	API review
AC-15	Two consecutive runs on identical inputs produce identical outputs.	BR-14, NFR-05	Repeat run
AC-16	No raw smart-meter data is present in the repository, and the capability runs without it.	BR-15	Inspection
AC-17	A deliberately degraded model is demonstrably blocked from reaching production by the quality gate.	BR-16	Negative test
AC-18	A failed release reverts automatically, leaving the previous model serving.	BR-16	Negative test
AC-19	The analysis refreshes on schedule without manual intervention.	BR-17	Observation

Ref	Criterion	Verifies	Method
AC-20	Every chart is accompanied by its data, and no status is conveyed by colour alone.	NFR-08	Accessibility review

Status at issue of this document

Criteria AC-01 through AC-16, AC-19 and AC-20 have been demonstrated on the reference implementation. AC-17 and AC-18 are exercised by automated negative tests of the release pipeline; their end-to-end demonstration against a live cloud endpoint is pending provision of the Azure subscription (dependency DP-02).

16. Benefits and Business Impact

16.1 Benefit summary

Table 16.23 - Anticipated benefits

Ref	Benefit	Type	Basis
BN-01	Better-timed reinforcement	Financial	Quantified exceedance replaces judgement in the timing decision, reducing both premature spend and late-constraint remediation.
BN-02	Better-targeted flexibility procurement	Financial	Stress is concentrated in identifiable half-hours and sites, so flexibility can be bought where it displaces a real constraint.
BN-03	Reduced analyst effort per study	Efficiency	A study that took weeks becomes a repeatable run measured in minutes, and can be refreshed whenever assumptions change.
BN-04	Stronger regulatory evidence base	Strategic	Investment cases supported by quantified, reproducible exceedance statistics and an explicit explanation of drivers.
BN-05	Earlier visibility of emerging constraints	Risk	Pre-emptive alerting on forecast rather than measured demand provides lead time.
BN-06	Transferable method	Strategic	The approach requires only smart-meter data and public context, so it extends to any area with equivalent data without new instrumentation.

16.2 Measured evidence from the reference implementation

Table 16.24 - Electrification impact measured across the adoption grid

Scenario	Peak amplification	Energy growth	Stress frequency	Escalation	Mean event duration
Base (no electrification)	—	—	4.4%	1.0×	2.2 h
EV 20%, heat pumps 15%	+52%	+29%	20.2%	4.6×	4.7 h
EV 40%, heat pumps 30%	+115%	+59%	31.8%	7.2×	5.5 h
EV 60%, heat pumps 50%	+190%	+93%	40.2%	9.1×	6.3 h
EV 80%, heat pumps 70%	+237%	+126%	46.8%	10.6×	6.9 h

The finding with the clearest commercial implication

Electric vehicles alone at 80% adoption raise the peak by 154%; heat pumps alone at 70% raise it by 113%. Together they raise it by 237% - **29 percentage points less than the sum of the two**. The loads peak at overlapping but not identical times, so treating them as additive overstates the combined constraint. Planning that adds separately-derived EV and heat-pump uplifts will therefore over-specify reinforcement.

16.3 Parameter credibility

Table 16.25 - Simulation assumptions checked against published planning practice

Quantity	Model output	Reference range
After-diversity maximum demand per EV	1.20 kW	Consistent with UK DNO planning figures for domestic charging.
After-diversity maximum demand per heat pump	1.30 kW	Consistent with UK DNO planning figures for domestic heat pumps.
Energy per vehicle per day	5.3 kWh	Approximately 1,900 kWh per year, consistent with average UK car mileage at typical efficiency.

These three quantities are reported by the system on every scenario run precisely so that a reviewer can perform this check independently, rather than accepting the headline figures on trust.

Caveat on the figures above

These results were produced on a calibrated reference dataset, not the licensed extract. They demonstrate that the capability computes what it claims to compute and that its assumptions are credible. They are **not** statements about any real London neighbourhood, and must not be quoted as such. Re-running the same pipeline on the licensed data is a configuration change, not a redevelopment.

17. Glossary

Table 17.26 - Business and technical terms used in this document

Term	Definition
ADMD After Diversity Maximum Demand	The peak electrical demand attributable to a single connected customer or device once the fact that not everyone uses power simultaneously is taken into account. The standard quantity UK distribution planners use for sizing.
Adoption elasticity	The marginal change in local peak demand produced by each additional tranche of technology uptake, expressed here per ten percentage points.
DNO	Distribution Network Operator - the licensed operator of a regional low- and medium-voltage electricity network.
Diversity factor	The reduction applied to a sum of individual peak demands to reflect that they do not all occur at the same instant.
Half-hour period	The settlement granularity of the GB electricity market; 48 periods per day.
Heating degrees	Degrees Celsius by which outside temperature falls below a reference balance point; the driver of electrified heating demand.
Low Carbon London	A smart-meter trial conducted in London, published through the UK Data Service as study 7857, providing half-hourly household consumption for 2011-2014.
LV feeder	Low-voltage circuit supplying a cluster of domestic properties from a distribution substation.
Neighbourhood	In this capability, a fixed-size group of smart-meter households used as the unit of analysis in place of a physical feeder, which the anonymised source data does not identify.
Peak amplification	The percentage increase in a neighbourhood's maximum half-hourly demand under an electrification scenario, relative to the un-electrified base case.
Pre-emptive alert	An alert raised because the <i>forecast</i> crosses the stress threshold, before measured demand has done so. The only alert type that provides lead time.
Promotion gate	The automated quality decision that determines whether a retrained model is allowed to replace the one currently in use.
Stress escalation	The multiple by which stress frequency increases under a scenario, relative to the base case.
Stress event	A contiguous run of two or more half-hours in which demand equals or exceeds the stress threshold. The unit in which duration statistics are reported.
Stress period	A single half-hour in which demand equals or exceeds the stress threshold.
Stress threshold	The 95th percentile of a neighbourhood's own historical half-hourly demand. A statistical reference point, not an asset rating.

18. Appendix A - Requirements Traceability Matrix

Every business requirement traces forward to the functional requirements that satisfy it and to the acceptance criteria that verify it. A requirement with no acceptance criterion cannot be signed off; a functional requirement with no business requirement is out of scope.

Table 18.27 - Traceability: business objective → business requirement → functional requirement → acceptance criterion

Objective	Business req.	Functional requirements	Acceptance
BO-1	BR-01	FR-01, FR-02, FR-03, FR-04	AC-01
BO-1	BR-02	FR-06, FR-07, FR-08, FR-09	AC-02, AC-03
BO-2	BR-03	FR-10, FR-11	AC-04, AC-11
BO-2	BR-04	FR-08, FR-12, FR-13, FR-25	AC-04
BO-3	BR-05	FR-14, FR-16, FR-17	AC-05, AC-07
BO-3	BR-06	FR-15, FR-16, FR-17	AC-06, AC-07
BO-3	BR-07	FR-11, FR-18	AC-05, AC-08
BO-4	BR-08	FR-19	AC-08
BO-5	BR-09	FR-21, FR-22	AC-09
BO-6	BR-10	FR-04	AC-12
BO-6	BR-11	FR-23, FR-24, FR-25	AC-10
BO-7	BR-12	FR-26, FR-27	AC-13, AC-20
BO-7	BR-13	FR-28	AC-14
BO-7	BR-14	FR-05, FR-09, FR-29	AC-01, AC-15
BO-8	BR-15	FR-03, FR-33	AC-16
BO-7	BR-16	FR-31, FR-32	AC-17, AC-18
BO-7	BR-17	FR-30	AC-19
BO-3	BR-18	FR-20	AC-05

Table 18.28 - Coverage check

Check	Result
Business requirements with at least one functional requirement	18 of 18
Business requirements with at least one acceptance criterion	18 of 18
Functional requirements traced to a business requirement	33 of 33
Business objectives covered by at least one requirement	8 of 8
Must-have requirements demonstrated at issue	All except the cloud-endpoint element of BR-16, pending DP-02

18.1 Requirements not carried forward

Table 18.29 - Requirements considered and deliberately excluded

Considered requirement	Reason for exclusion
Stress defined relative to transformer rating	The available data contains no asset ratings. Adopting an assumed rating would have produced authoritative-looking figures resting on an invented constant. Recorded as constraint CN-03 and limitation BX-01.
Automatic reinforcement recommendation	Follows from the above. The capability informs the decision; it cannot make it.
Household-level insight for targeted engagement	Prohibited by the data licence and unnecessary for the business purpose.
Smart-tariff and managed-charging scenarios	Excluded from this phase so that results characterise the uncontrolled worst case. Carried forward as BF-01.